Fiscal Year:	FY 2008	**Task Last Updated:**	FY 06/29/2009
PI Name:	Pitz, Robert Wendell Ph.D.		
Project Title:	CURVED LAMINAR FLAMES IN MICROGRAVITY		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Nashville	**State:**	TN
Zip Code:	37235-0001	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	10/01/2003	**End Date:**	09/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2008 per PI and notice from GRC (6/09) NOTE: Received NCE to 9/30/2007, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04AA14A		
Performance Goal No.:			
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Task Description:	A cylindrical burner will be studied under microgravity conditions in both the 2.2 second drop tower and the aircraft KC135. Diagnostic techniques under microgravity condition will include color video, (CH* and OH*) chemiluminescent imaging, and the Rainbow Schlieren method. Normal gravity investigations will also be performed at Vanderbilt University and will include detailed measurements of flame structure using combustion laser diagnostics. Three different kinds of fuels will be investigated in order to examine the effects of varying Lewis number. Task Objective: 1. Examine the effects of curvature on the flame structure of laminar diffusion flames. 2. Investigate the effects of curvature on the extinction limit of laminar diffusion flames. 3. Improve either reaction rate data or transport data for flames with a high degree of curvature. Practical combustion devices inevitably involve turbulence to some extent, which is characterized by the concurrence of stretch and curvature. In certain circumstances, curvature may be the dominant factor of some combustion phenomena, taking the extinction of the weak turbulent diffusion flame as an example. The unique geometry of the Vanderbilt cylindrical burner enables the separation of stretch effects and curvature effects, and therefore makes the investigation of these combustion phenomena possible. A comparison of the extinction stretch rate between counterflow flames and tubular flames can reveal the effects of curvature on the extinction limit. Assisted by numerical investigations, this result will reveal improved ways to prevent or even extinguish fire under microgravity. New insights to geometry-related fire safety concerns might also be introduced. A recent numerical study showed the non-monotonous dependence on the stretch rate of the temperature of a negatively curved laminar flame with Lewis number less than one. Such phenomena can only be seen at very low stretch rate, where the gravity effects will compromise or even prevent the discovery of this phenomenon. Only under microgravity condition, such a flame can be established and thus make the study of such kind of curvature effects feasible.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understand effect of flame curvature on flame stability and extinction which will aid in control of fires.
Task Progress:	The effect of curvature on laminar flames is studied in tubular flames. For non-premixed tubular flames, curvature effects are best studied in micro-gravity to avoid deleterious effects of buoyancy. For micro-gravity study, a drop tube experiment was developed around the Vanderbilt opposed tubular burner. The drop tube experiment included fuel/oxidizer, ignition, instrumentation, and data acquisition systems. The drop tube rig passed the design review in the first 9 months of the project and the rig was constructed by NASA and Vanderbilt over the next 15 months. Prior to the safety review at the start of the 3rd year of the project, NASA re-directed its research programs and the drop tube was no longer available as a government provided facility. Extensive experiments on the effect of curvature on laminar flames in normal gravity were performed using the opposed flow tubular burner. In flame structure studies, laser-induced Raman scattering was applied to measure the temperature and major chemical species concentrations in non-premixed opposed tubular flames burning hydrogen, methane and propane fuels. The measured temperature and chemical species concentration profiles in the tubular flames were compared to detailed transport, complex chemistry numerical simulations; good agreement was found between the measured and predicted results. The importance of curvature on the temperature and extinction of non-premixed flames was found to depend on the fuel Lewis number, Lef, that is the ratio of the bulk thermal diffusivity to the fuel mass diffusivity. For example, when compared to flat flames, for opposed tubular flames burning 15% hydrogen diluted in nitrogen (Lef=0.41; a value less than 1), convex curvature toward the fuel increased the flame temperature by 80K and the extinction stretch rate was 20% higher; the flame with concave curvature toward the fuel decreased the flame temperature by 80K and the extinction stretch rate was 50% lower. The opposite effect was found for flames with fuel Lewis numbers above one. The effect of curvature is greatest for sub-unity fuel Lewis number hydrogen flames. The effect of flame curvature on extinction on non-premixed flames for hydrogen, methane and hydrogen fuels diluted by various inert gases (Ar, He, N2, CO2) burning in air was studied in the opposed flow tubular burner. For fuel mixtures with fuel Lewis number much less than one, convex flame curvature made the flame more difficult to extinguish (vice versa for concave curvature). The opposite effect was found for fuel mixtures with fuel Lewis numbers at or above one. Thus curvature can make flames harder to extinguish. The cellular regions of non-premixed tubular flames for low fuel Lewis number hydrogen diluted in inert gas (He, N2, Ar, CO2) vs. air flames were mapped out. Two to eight cells were found along the circumference of the diluted H2 flames near extinction. Convex curvature promoted the cellular behavior and concave curvature suppressed the cellular behavior. These instabilities were due to thermal-diffusive effects since they were not found in near unity fuel Lewis number flames. The cellular structure of premixed tubular flames was also studied.
Bibliography Type:	Description: (Last Updated: 02/18/2011)
Articles in Peer-reviewed Journals	Hu S, Wang P, Pitz RW, Smooke MD. "Experimental and numerical investigation of non-premixed tubular flames." Proceedings of the Combustion Institute. 2007 Jan;31(1):1093-9. http://dx.doi.org/10.1016/j.proci.2006.08.058 , Jan-2007
Articles in Peer-reviewed Journals	Hu S, Pitz RW. "Structural study of non-premixed tubular hydrocarbon flames." Combustion and Flame. 2009 Jan;156(1):51-61. http://dx.doi.org/10.1016/j.combustflame.2008.07.017 , Jan-2009
Articles in Peer-reviewed Journals	Hu S, Pitz RW, Wang Y. "Extinction and near-extinction instability of non-premixed tubular flames." Combustion and Flame. 2009 Jan;156(1):90-8. http://dx.doi.org/10.1016/j.combustflame.2008.09.004 , Jan-2009
Articles in Peer-reviewed Journals	Hu S, Wang P, Pitz RW. "A structural study of premixed tubular flames." Proceedings of the Combustion Institute. 2009;32(1):1133-40. http://dx.doi.org/10.1016/j.proci.2008.06.183 , Jan-2009
Articles in Peer-reviewed Journals	Wang Y, Hu S, Pitz RW. "Extinction and cellular instability of premixed tubular flames." Proceedings of the Combustion Institute. 2009;32(1):1141-7. http://dx.doi.org/10.1016/j.proci.2008.07.012 , Jan-2009

Papers from Meeting Proceedings	Hu S, Wang P, Pitz RW. "A structural study of premixed tubular flames." 44th AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, January 9-12, 2006. Washington, DC : American Institute of Aeronautics and Astronautics, 2006. Paper AIAA-2006-0160. , Jan-2006
Papers from Meeting Proceedings	Hu S, Wang P, Pitz RW, Smooke MD. "Curvature effects in non-premixed tubular flames." 44th AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, January 9-12, 2006. Washington, DC : American Institute of Aeronautics and Astronautics, 2006. Paper AIAA-2006-1133. , Jan-2006
Papers from Meeting Proceedings	Hu S, Pitz RW, Wang Y. "Extinction and near extinction instability of non-premixed tubular flames." 45th AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, January 8-11, 2007. Washington, DC : American Institute of Aeronautics and Astronautics, 2007. Paper AIAA-2007-0177. , Jan-2007
Papers from Meeting Proceedings	Hu S, Pitz RW. "Structural study of non-premixed tubular hydrocarbon flames." 45th AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, January 8-11, 2007. Washington, DC : American Institute of Aeronautics and Astronautics, 2007. Paper AIAA-2007-0178. , Jan-2007
Papers from Meeting Proceedings	Wang Y, Hu S, Pitz RW. "Cellular instabilities in non-premixed opposed-flow tubular flames." 47th AIAA Aerospace Sciences Meeting and Exhibit, Orlando, Florida, January 4-8, 2009. Washington, DC : American Institute of Aeronautics and Astronautics, 2009. Paper AIAA-2009-1181. , Jan-2009